

Lezione 1 Studi di funzione

Matematica del continuo

Modulo 11 - Studio di funzioni

Unità didattica 4 – Funzioni iperboliche

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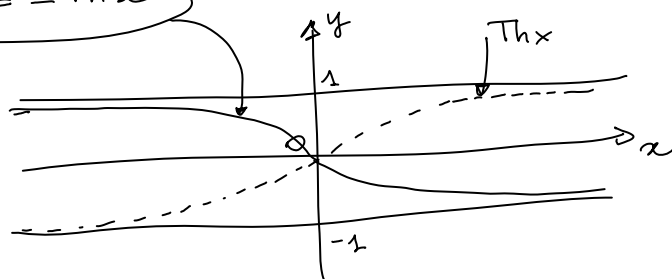
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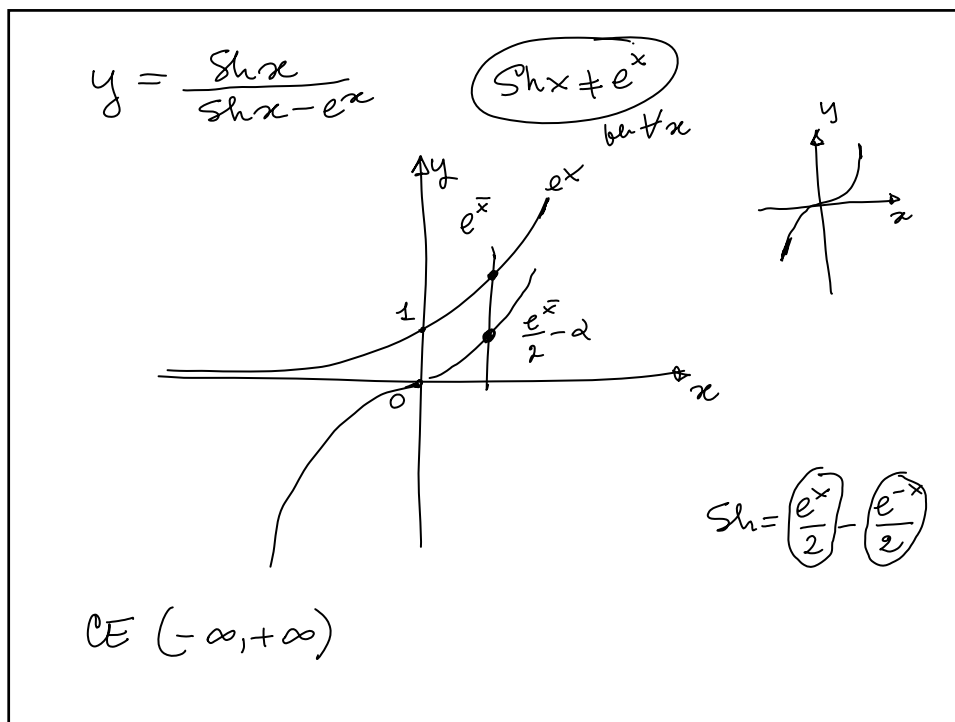
$$y = \frac{\operatorname{Sh} x}{\operatorname{Sh} x - e^x}$$

$$\operatorname{Sh} x = \frac{e^x - e^{-x}}{2}$$

$$y = \frac{e^x - e^{-x}}{2} \cdot \frac{x}{e^x - e^{-x} - 2e^x} = -\frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$y = -\operatorname{Th} x$$





$y = \frac{\operatorname{Sh} x}{\operatorname{Sh} x - e^x} \quad \text{CE } (-\infty, +\infty)$

$(0,0)$

$\lim_{x \rightarrow -\infty} y = 1 \quad \frac{\operatorname{Sh}(-\infty)}{\operatorname{Sh}(-\infty) - e^{-\infty}}$

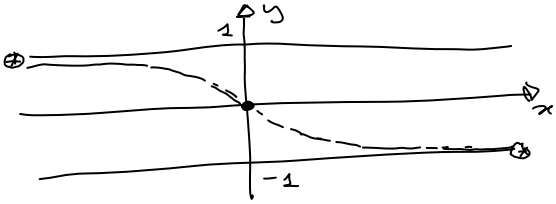
$y=1$ As. Orizz. a $-\infty$

$\lim_{x \rightarrow +\infty} y = -1 \quad \frac{\operatorname{Sh}(+\infty)}{\operatorname{Sh}(+\infty) - e^{+\infty}}$

$\frac{\frac{e^x - e^{-x}}{2}}{\frac{e^x - e^{-x}}{2} - e^x} = \frac{\cancel{e^x} e^{-x}}{2} \cdot \frac{2}{-\cancel{e^x} + e^{-x}}$

$y = -1$
 As. Orizz.
 b/w $x \rightarrow +\infty$

$$y = \frac{\sinh x}{\sinh x - e^x}$$



$$y' = \frac{1}{(\sinh x - e^x)^2} (\cosh x (\sinh x - e^x) - \sinh x (\cosh x - e^x))$$

$$= \frac{(-1)}{(\sinh x - e^x)^2} e^x (\underbrace{\cosh x - \sinh x}_{>0})$$

$y' < 0 \quad \text{for } x \in \mathbb{R}$

$y \text{ decrease}$

$y'' = \dots$

$y'(0) = ?$

$\uparrow ?$

